Orbital Stability of the Complete Minimizing Set for a Fourth-Order Cubic–Quintic Schrödinger Equation

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Abstract

For the Hamiltonian flow $i\partial_t u = (\Delta + k^2)^2 u - \beta|u|^2 u + \gamma|u|^4 u$ on $\mathbb{R}^2$, with $\beta, \gamma, k > 0$, we give a complete variational proof that the full mass-constrained minimizing set is orbitally stable in H^2 for $0 < m \le m_B$, where $m_B = 3\beta/[\gamma\sqrt{k^4 + 3\beta^2/(32\gamma)}]$. The argument proves negativity of the infimum, strict binding, compactness modulo translations, and global well-posedness, then applies conservation of mass and energy. Perturbations are not required to preserve mass exactly. This concerns the entire minimizing set, without uniqueness, attraction, or spectral nondegeneracy.

1 Statement and notation

Write $X = H^2(\mathbb{R}^2;\mathbb{C})$, $A = \Delta + k^2$, and

$$M(u) = \|u\|_2^2, \qquad D(u) = \|\Delta u\|_2^2, \qquad K(u) = \|\nabla u\|_2^2,$$
$$T(u) = \|Au\|_2^2 = D(u) - 2k^2 K(u) + k^4 M(u), \qquad Q(u) = \|u\|_4^4, \qquad S(u) = \|u\|_6^6.$$

Set

$$H(u) = T(u) - \frac{\beta}{2}Q(u) + \frac{\gamma}{3}S(u), \qquad I(m) = \inf_{M(u)=m} H(u),$$
$$\mathcal{G}_m = \{u \in X : M(u) = m,\ H(u) = I(m)\}.$$

Let

$$\rho = k^4 + \frac{3\beta^2}{32\gamma}, \qquad \sigma = \sqrt{\rho}, \qquad m_B = \frac{3\beta}{\gamma\sigma}.$$

For a nonempty subset $E \subset X$, define $\operatorname{dist}_X(v, E) = \inf_{g\in E}\|v - g\|_{H^2}$.

Theorem 1 (Existence, compactness, and full-set orbital stability)**.** *For each $0 < m \le m_B$:*

(i) $-\infty < I(m) < 0$, and $I(m) < I(a) + I(m-a)$ for all $0 < a < m$;

(ii) $\mathcal{G}_m \neq \varnothing$, and every mass-$m$ minimizing sequence has a subsequence converging strongly in H^2 after spatial translations to an element of $\mathcal{G}_m$;

(iii) every datum in X generates a unique global solution $u \in C(\mathbb{R}; X)$, conserving both M and H;

(iv) for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\operatorname{dist}_X(u_0, \mathcal{G}_m) < \delta \quad \Longrightarrow \quad \sup_{t\in\mathbb{R}} \operatorname{dist}_X(u(t), \mathcal{G}_m) < \varepsilon.$$

The conclusion in (iv) permits initial data with $M(u_0) \neq m$.

2 Coercivity and strict negativity

Lemma 2 (Coercivity at fixed mass)**.** *With $C_0 = 3\beta^2/(16\gamma)$, every $u \in X$ satisfies*

$$H(u) \geq T(u) - C_0 M(u), \qquad \|\Delta u\|_2 \leq \sqrt{H(u) + C_0 M(u)} + k^2\sqrt{M(u)}.$$

On a set of bounded mass and bounded-above energy, the H^2 norm is uniformly bounded.

Proof. For $r \geq 0$, completion of the square gives

$$-\frac{\beta}{2}r^2 + \frac{\gamma}{3}r^3 = \frac{\gamma}{3}r\left(r - \frac{3\beta}{4\gamma}\right)^2 - \frac{3\beta^2}{16\gamma}r.$$

Integrating with $r = |u|^2$ gives the first inequality. Since $\Delta u = Au - k^2 u$, the second follows from the triangle inequality. Finally $\|\nabla u\|_2^2 \leq \|u\|_2 \|\Delta u\|_2$ by Fourier Cauchy–Schwarz, so the L^2 and Laplacian bounds control $\|u\|_{H^2}$. □

Lemma 3 (Strict negativity at every positive mass)**.** *For every $m > 0$, $-C_0 m \leq I(m) < 0$.*

Proof. The lower bound is Lemma 2. For $a, b, R > 0$, take

$$f_{a,b}(x,y) = (\pi ab)^{-1/2} \exp\left(-\frac{x^2}{2a^2} - \frac{y^2}{2b^2}\right), \qquad u_R(x,y) = \frac{\sqrt{m}}{R} f_{a,b}(x/R, y/R) e^{ikx}.$$

Then $M(u_R) = m$. A direct Gaussian calculation, using $A(e^{ikx}f) = e^{ikx}(\Delta f + 2ik\partial_x f)$, yields

$$H(u_R) = \frac{c_2}{R^2} + \frac{c_4}{R^4},$$

where

$$c_2 = \frac{2mk^2}{a^2} - \frac{\beta m^2}{4\pi ab},$$
$$c_4 = m\left(\frac{3}{4a^4} + \frac{1}{2a^2b^2} + \frac{3}{4b^4}\right) + \frac{\gamma m^3}{9\pi^2 a^2 b^2} > 0.$$

Choosing $a/b > 8\pi k^2/(\beta m)$ makes $c_2 < 0$. The choice $R^2 = -2c_4/c_2$ then gives $H(u_R) = -c_2^2/(4c_4) < 0$. Thus $I(m) < 0$. □

3 Energy per mass and strict binding

The dilation $u_\lambda(x) = \lambda u(\lambda x)$ preserves mass *in two dimensions*. Its energy is

$$H(u_\lambda) = \lambda^4\left(D(u) + \frac{\gamma}{3}S(u)\right) - \lambda^2\left(2k^2K(u) + \frac{\beta}{2}Q(u)\right) + k^4 M(u). \tag{1}$$

Lemma 4 (Exact dilation adjustment)**.** *Any minimizing sequence (u_n) at any mass $a > 0$ can be replaced by another minimizing sequence (w_n) at the same mass such that*

$$D(w_n) - k^2 K(w_n) - \frac{\beta}{4}Q(w_n) + \frac{\gamma}{3}S(w_n) = 0 \tag{2}$$

for every n.

Proof. Put $P = D(u) + \gamma S(u)/3 > 0$ and $B = 2k^2K(u) + \beta Q(u)/2 > 0$ for $u \neq 0$. Equation (1) is $Pz^2 - Bz + k^4 a$ with $z = \lambda^2 > 0$; its minimum occurs at $z = B/(2P) > 0$. Set $w_n = (u_n)_{\lambda_n}$ with this minimizing scale. Then $I(a) \le H(w_n) \le H(u_n) \to I(a)$. Differentiation of the exact polynomial at $\lambda = 1$, now applied to each adjusted profile, yields (2). No differentiability of the map $\lambda \mapsto u_\lambda$ in H^2 is assumed. □

Lemma 5 (Uniform nonlinear ratio at an adjusted profile)**.** *If $w \in X \setminus \{0\}$ satisfies (2), then*

$$D(w) \le \rho M(w), \qquad \frac{S(w)}{Q(w)} \le \frac{\sigma M(w)}{4}.$$

Proof. Writing $r = |w|^2$, substitution of (2) into the identity $T = D - 2k^2K + k^4M$ gives the exact equality

$$D(w) + T(w) + \frac{2\gamma}{3}\int_{\mathbb{R}^2} r\left(r - \frac{3\beta}{8\gamma}\right)^2 dx = \left(k^4 + \frac{3\beta^2}{32\gamma}\right) M(w) = \rho M(w). \tag{3}$$

All three terms on the left are nonnegative, so $D(w) \le \rho M(w)$. With the Fourier convention $f(x) = (2\pi)^{-2}\int e^{ix\cdot\xi}\widehat{f}(\xi)\,d\xi$, Cauchy–Schwarz and polar integration show, for any $\eta > 0$,

$$\|w\|_\infty^2 \le \frac{D(w) + \eta M(w)}{8\sqrt{\eta}}, \qquad \frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} \frac{d\xi}{|\xi|^4 + \eta} = \frac{1}{8\sqrt{\eta}}.$$

Here $D(w) > 0$ for nonzero $w \in L^2$. Taking $\eta = D(w)/M(w)$ yields $\|w\|_\infty^2 \le \sqrt{M(w)D(w)}/4 \le \sigma M(w)/4$. Finally $S(w) \le \|w\|_\infty^2 Q(w)$, with $Q(w) > 0$. □

Lemma 6 (Strictly decreasing specific energy)**.** *Let $e(a) = I(a)/a$. If $0 < a < b$ and $a + b < 2m_B$, then*

$$e(b) \le \left[1 + \frac{b-a}{a}\left(1 - \frac{a+b}{2m_B}\right)\right] e(a) < e(a). \tag{4}$$

Consequently e is strictly decreasing on $(0, m_B]$.

Proof. Put $t = b/a > 1$, $h = 1 - (a+b)/(2m_B) > 0$, and choose an adjusted mass-a minimizing sequence (w_n) from Lemma 4. Lemma 5 gives

$$\frac{\gamma}{3}(t+1)S(w_n) \le \frac{\gamma(t+1)\sigma a}{12} Q(w_n) = \frac{\beta}{2}\frac{a+b}{2m_B} Q(w_n).$$

Since $M(\sqrt{t}\, w_n) = b$, the amplitude-scaling identity yields

$$\begin{aligned}
\frac{I(b)}{t} &\le \frac{H(\sqrt{t}\, w_n)}{t} \\
&= H(w_n) - (t-1)\left(\frac{\beta}{2}Q(w_n) - \frac{\gamma}{3}(t+1)S(w_n)\right) \\
&\le H(w_n) - (t-1)\frac{\beta h}{2} Q(w_n).
\end{aligned}$$

Because $T \ge 0$ and $S \ge 0$, one has $Q(w_n) \ge -2H(w_n)/\beta$. Therefore

$$\frac{I(b)}{t} \le [1 + (t-1)h] H(w_n).$$

Taking $n \to \infty$ and dividing by a proves the first inequality in (4). The second is strict because $e(a) = I(a)/a < 0$ by Lemma 3 and $(b-a)h/a > 0$. If $0 < a < b \le m_B$, then $a + b < 2m_B$, so the last assertion follows. □

Proposition 7 (Strict binding)**.** *For every $0 < m \le m_B$ and every $0 < a < m$,*

$$I(m) < I(a) + I(m-a).$$

Proof. By Lemma 6, $e(a) > e(m)$ and $e(m-a) > e(m)$. Multiplying by the positive masses and adding,

$$I(a) + I(m-a) = ae(a) + (m-a)e(m-a) > me(m) = I(m).$$

□

4 Compactness modulo translations

Proposition 8 (Strong compactness of minimizing sequences)**.** *Fix $0 < m \le m_B$. If $M(u_n) = m$ and $H(u_n) \to I(m)$, then there exist a subsequence, translations $y_n \in \mathbb{R}^2$, and $g \in \mathcal{G}_m$ such that $u_n(\cdot + y_n) \to g$ strongly in $H^2(\mathbb{R}^2)$.*

Proof. By Lemma 2, (u_n) is bounded in H^2. We first rule out *vanishing*, namely

$$\sup_{y \in \mathbb{R}^2} \int_{B(y,1)} |u_n|^2 \, dx \longrightarrow 0.$$

Cover $\mathbb{R}^2$ by unit squares Q_j (each contained in a unit ball after translation). The local two-dimensional Gagliardo–Nirenberg estimate on a unit square states

$$\int_{Q_j} |v|^4 \, dx \le C \left(\int_{Q_j} |v|^2 \, dx \right) \left(\int_{Q_j} (|\nabla v|^2 + |v|^2) \, dx \right).$$

Summation over j would give $Q(u_n) \le C(\sup_j \|u_n\|^2_{L^2(Q_j)}) \|u_n\|^2_{H^1} \to 0$. As $H^2(\mathbb{R}^2) \hookrightarrow L^\infty$, we also get $S(u_n) \le \|u_n\|^2_\infty Q(u_n) \to 0$. This would imply $I(m) = \lim H(u_n) \ge 0$, contrary to Lemma 3.

Thus suitable translations make $v_n = u_n(\cdot + y_n)$ retain a uniformly positive L^2 mass in a fixed ball. Passing to a subsequence, $v_n \rightharpoonup v$ weakly in H^2, strongly in L^p_{loc} for all finite p, and almost everywhere. The local mass lower bound implies $v \neq 0$. Put $a = M(v) \in (0, m]$ and $r_n = v_n - v$. Weak convergence in H^2 and the Hilbert-space identities give

$$M(r_n) = m - a + o(1), \qquad T(v_n) = T(v) + T(r_n) + o(1). \tag{5}$$

The Brezis–Lieb lemma, applicable at powers 4 and 6 by boundedness and almost-everywhere convergence, gives

$$Q(v_n) = Q(v) + Q(r_n) + o(1), \qquad S(v_n) = S(v) + S(r_n) + o(1).$$

Hence

$$H(v_n) = H(v) + H(r_n) + o(1). \tag{6}$$

If $a < m$, let $b = m - a > 0$ and $\widetilde{r}_n = \sqrt{b/M(r_n)}\, r_n$. The multiplying factor converges to 1, and (r_n) is uniformly bounded in H^2. Since H is a continuous polynomial functional with locally bounded derivative on H^2, $H(\widetilde{r}_n) - H(r_n) \to 0$. Thus

$$I(m) = \lim H(v_n) = H(v) + \lim H(r_n) \ge I(a) + I(b),$$

contradicting Proposition 7. Therefore $a = m$.

Now $M(r_n) \to 0$. The uniform H^2 bound implies $Q(r_n)+S(r_n) \to 0$, since $Q(r_n) \le \|r_n\|_\infty^2 M(r_n)$ and $S(r_n) \le \|r_n\|_\infty^4 M(r_n)$. Equation (6) becomes $I(m) = H(v) + \lim T(r_n)$, while $H(v) \ge I(m)$ and $T(r_n) \ge 0$. It follows that $H(v) = I(m)$ and $T(r_n) \to 0$. Finally,

$$\|\Delta r_n\|_2 \le \|Ar_n\|_2 + k^2\|r_n\|_2 \to 0, \quad \|\nabla r_n\|_2^2 \le \|r_n\|_2\|\Delta r_n\|_2 \to 0.$$

Thus $r_n \to 0$ strongly in H^2, and $v \in \mathcal{G}_m$. □

Remark 9. *The argument does not assume a radial, periodic, or real-valued class. Translations are essential: $\mathcal{G}_m$ is not generally compact in the original coordinates.*

5 The global Hamiltonian flow

Proposition 10 (Global well-posedness and conservation)**.** *For each $u_0 \in X$, the Cauchy problem*

$$i\partial_t u = A^2 u - \beta|u|^2 u + \gamma|u|^4 u, \qquad u(0) = u_0 \tag{7}$$

has a unique $u \in C(\mathbb{R}; X)$, depends continuously on data on bounded time intervals, and satisfies $M(u(t)) = M(u_0)$ and $H(u(t)) = H(u_0)$ for all $t \in \mathbb{R}$.

Proof. The symbol of A^2 is the real function $(k^2 - |\xi|^2)^2$. Therefore $U(t) = e^{-itA^2}$ is a strongly continuous unitary group on every H^s. In dimension two, $H^2 \hookrightarrow L^\infty$ and H^2 is a Banach algebra. Consequently the map $F(u) = -\beta|u|^2u + \gamma|u|^4u$ is locally Lipschitz $X \to X$. The Duhamel equation

$$u(t) = U(t)u_0 - i\int_0^t U(t-s)F(u(s))\,ds$$

has, by contraction on $C([-\tau,\tau]; X)$ for small enough τ, a unique maximal solution and the usual blow-up alternative: if a finite positive or negative endpoint of the lifespan exists, $\|u(t)\|_{H^2}$ becomes unbounded there.

For completeness, conservation is justified at the energy regularity. If $u_0 \in H^4$, the same contraction gives an H^4 solution, and $u_t \in L^2$. Pairing (7) with u and taking real parts shows $dM(u(t))/dt = 0$, because A^2 is self-adjoint and both nonlinear pairings are real. The real Fréchet derivative of H obeys

$$DH(u)[h] = 2\,\mathrm{Re}\int_{\mathbb{R}^2} \overline{\left(A^2u - \beta|u|^2u + \gamma|u|^4u\right)}\, h\,dx.$$

Thus $dH(u(t))/dt = 2\,\mathrm{Re}\langle iu_t, u_t\rangle = 0$. (One may first use smoother data and pass to H^4.)

For general $u_0 \in X$, choose $u_{0,j} \in H^4$ converging in H^2. On any compact interval contained in the maximal H^2 lifespan, the local continuous-dependence estimates imply convergence of the corresponding smooth solutions in $C_tH^2_x$. Smooth solutions continue in H^4 throughout such an interval: the standard tame product bound $\|F(u)\|_{H^4} \le C(\|u\|_{H^2})\|u\|_{H^4}$ and Gronwall's inequality prevent an H^4 blow-up while H^2 is bounded. Since M and H are continuous on H^2, their conservation passes to the limit.

Finally, Lemma 2 and the conservation laws give

$$\|Au(t)\|_2^2 \le H(u_0) + C_0M(u_0), \qquad \|\Delta u(t)\|_2 \le \sqrt{H(u_0) + C_0M(u_0)} + k^2\sqrt{M(u_0)}.$$

Together with fixed L^2 mass, these yield a time-independent H^2 bound. The blow-up alternative excludes both finite-time endpoints; hence the solution exists for all $t \in \mathbb{R}$. □

6 Orbital stability of the full minimizing set

Proof of Theorem 1(iv). Fix $m \in (0, m_B]$. The first two claims of the theorem were proved in Lemma 3, Proposition 7, and Proposition 8; the third is Proposition 10. Moreover $\mathcal{G}_m$ is invariant under the global flow by conservation of mass and energy, and is invariant under spatial translations by the translation invariance of M and H.

First note that $\mathcal{G}_m$ is uniformly bounded in H^2: for any $g \in \mathcal{G}_m$, Lemma 2 gives $T(g) \leq I(m) + C_0 m$, so $\|g\|_{H^2} \leq C(m, \beta, \gamma, k)$.

Suppose stability fails. There are $\varepsilon_0 > 0$ and, for all sufficiently large n, data $u_{0,n} \in X$ for which $\operatorname{dist}_X(u_{0,n}, \mathcal{G}_m) < 1/n$ while the supremum of the distance along the flow is at least ε_0. For each n, select $t_n \in \mathbb{R}$ with distance at least $\varepsilon_0/2$; this does not require the supremum to be attained. Then

$$\operatorname{dist}_X(u_{0,n}, \mathcal{G}_m) \to 0, \qquad \operatorname{dist}_X(u_n(t_n), \mathcal{G}_m) \geq \varepsilon_0/2, \tag{8}$$

where $u_n(t)$ is the global solution launched from $u_{0,n}$. Choose $g_n \in \mathcal{G}_m$ with $\|u_{0,n} - g_n\|_{H^2} \to 0$. By the uniform bound on g_n, $(u_{0,n})$ is bounded in H^2. Continuity of mass and energy, uniform on bounded H^2 sets for these polynomial functionals, yields

$$M(u_{0,n}) \to m, \qquad H(u_{0,n}) \to I(m). \tag{9}$$

Set $v_n = u_n(t_n)$. Conservation transfers (9) to $M(v_n) \to m$ and $H(v_n) \to I(m)$. Lemma 2 ensures (v_n) is bounded in H^2. For large n, define the *mass-normalized* functions

$$\alpha_n = \sqrt{\frac{m}{M(v_n)}}, \qquad w_n = \alpha_n v_n.$$

Then $\alpha_n \to 1$, $M(w_n) = m$, and

$$\|w_n - v_n\|_{H^2} = |\alpha_n - 1| \|v_n\|_{H^2} \to 0.$$

The polynomial form of H and boundedness in H^2 imply $H(w_n) - H(v_n) \to 0$, so

$$M(w_n) = m, \qquad H(w_n) \to I(m).$$

By Proposition 8, after passing to a subsequence there exist $y_n \in \mathbb{R}^2$ and $g \in \mathcal{G}_m$ such that $w_n(\cdot + y_n) \to g$ strongly in H^2. Equivalently, $\|w_n - g(\cdot - y_n)\|_{H^2} \to 0$. As $g(\cdot - y_n) \in \mathcal{G}_m$, we obtain

$$\operatorname{dist}_X(v_n, \mathcal{G}_m) \leq \|v_n - w_n\|_{H^2} + \|w_n - g(\cdot - y_n)\|_{H^2} \longrightarrow 0.$$

This contradicts (8). Therefore orbital stability holds for every $\varepsilon > 0$, including mass-changing perturbations. □

7 Interpretation and antecedents

The conclusion is *Lyapunov stability of a set*. It does not imply convergence as $|t| \to \infty$, uniqueness of the minimizing profile, stability of an arbitrary stationary solution outside $\mathcal{G}_m$, or spectral nondegeneracy. The proof is specifically for two spatial dimensions and does not classify four-spatial-dimensional lifetimes.

The final conservation-plus-compactness step follows the classical Cazenave–Lions variational stability principle. Its application here requires the preceding specific binding and compactness estimates for the mixed cubic–quintic Hamiltonian. Closely related fourth-order single-power mixed-dispersion problems were treated in the papers listed below.

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